

WASEEM ALMAZRUA

Machine Learning Engineer | AI Engineer | Computer Vision & AI Systems
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SUMMARY

Result-oriented Machine Learning Engineer and AI Engineer bridging advanced computer vision and production-grade AI systems. Leveraging a clinical background from KFMC to architect high-precision medical image analysis and cross-industry AI pipelines — turning domain expertise into a technical advantage. Proven track record in full-stack AI deployment, microservices architectures using FastAPI and BentoML, and scalable cloud infrastructure on GCP with strict MLOps practices including model versioning, CI/CD, experiment tracking, and containerized deployment.

SKILLS

Languages:	Python, SQL
Libraries:	pandas, NumPy, scikit-learn, Matplotlib, TensorFlow, Keras, PyTorch, OpenCV, transformers, MONAI, Pydantic, Alembic
Frameworks:	FastAPI, Pydantic AI
Tools:	Docker, MLflow, BentoML, CI/CD, Linux, Git, Power BI, PostgreSQL
Platforms:	Google Cloud, GitHub, DigitalOcean, Hugging Face
Domains:	Computer Vision, Deep Learning, Model Deployment, Inference Optimization, Medical Image Analysis, MLOps, Model Serving, Model Monitoring, Production ML, Agentic AI, AI Agents, RAG, Vector Databases, Embeddings, NLP, LLM Integration

PROJECTS

Doctor Copilot — Advanced Brain MRI Segmentation AI Link	2025
Deployed Double U-Net + ResNet50 model for brain MRI segmentation achieving Dice 0.87 and IoU 0.82; built a REST API via FastAPI on Google Cloud Run and integrated OpenAI API to power an AI chatbot for clinical result interpretation.	
Technologies: Python, TensorFlow, FastAPI, Google Cloud Run, Docker, U-Net, ResNet50	
Face Mask Detection System — YOLOv8 Link	2025
Built a real-time YOLOv8 system classifying mask compliance (with/without/incorrect) in hospital environments, deployed on Google Cloud Run.	
Technologies: YOLOv8, PyTorch, OpenCV, FastAPI, Google Cloud Run	
PPE Monitoring System — Computer Vision Link	2025
Real-time PPE detection system using deep learning, achieving mAP@0.5 of 94%; supports live video compliance flagging in hospital and industrial settings.	
Technologies: Python, PyTorch, YOLO, OpenCV, Deep Learning, Computer Vision	
Brats-3D — Brain Tumor Segmentation Platform Link	2026
Full-stack medical AI platform for 3D brain MRI segmentation: React + Auth0 frontend, FastAPI + BentoML microservices backend on Google Cloud Run, model versioning via MLflow model registry with PyFunc wrapper and BentoML registry, direct-to-GCS uploads via signed URLs, automated CI/CD with GitHub Actions, and OpenAI API integration to generate AI-powered clinical summaries of segmentation results.	
Technologies: Python, BentoML, FastAPI, React, Google Cloud Run, GCS, Auth0, GitHub Actions, PostgreSQL, Docker	

EDUCATION

Bachelor's Degree in Nursing Science	2017 – 2022
Shaqra University GPA: 4.88 / 5.0	

Data Science & AI Bootcamp	Aug – Nov 2025
SDA x Le Wagon — Intensive 12-week program	
Covered Python, SQL, machine learning, deep learning, data engineering, and real-world AI project deployment.	

CERTIFICATIONS

Data Analyst Nanodegree — Udacity AI Programming with Python — Udacity Programming for Data Science with Python — Udacity Statistics for Data Analysis — Udacity
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EXPERIENCE

Staff Nurse — Surgical Unit	Oct 2022 – Oct 2024
King Fahad Medical City (KFMC) Riyadh, Saudi Arabia	
- Managed care delivery for 4+ surgical patients per shift, maintaining patient safety and clinical documentation standards. - Collaborated with multidisciplinary teams to support clinical quality improvement and outcome tracking initiatives.	